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ABO Blood Type and Personality Traits: Evidence from Large-scale Surveys in Japan

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The relationship between blood type and personality has long been one of the more challenging issues of scientific studies. Several large-scale surveys were conducted to address the issue, and some of them had shown statistically significant associations. This study analyzed data from two large-scale surveys (Survey 1: N = 1,000, Survey 2: N = 1,859) to examine the relationship between blood type and personality. ANOVA results indicated that 17 of the total 20 respondents' own blood type characteristic question items scored higher as "fit to my personality" than the averages of the other blood types. In both Survey 1 and 2, the same differences in scores were found in the groups who reported no blood type personality knowledge, although the values were smaller. Thus, we observed a clear and significant relationship between blood type and personality in large-scale surveys.

Keywords

ai, big five, blood type, large-scale survey, personality

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Keywords

Blood type, personality, large-scale survey

Introduction

Background

Currently, the effect of genetic factors on personality is estimated to be approximately 50% (Ando et al. 2004; Gerra et al., 2000, Keller et al., 2005). The ABO blood group, discovered by Karl Landsteiner in 1901, is a genetically determined and easily measured biological marker. Hence, much research has been conducted not only on health risks, but also on its impact on personality.

The relationship between blood type and personality is studied at an international scale, and the first academic one using statistics a base was conducted in 1927, by Takeji Furukawa, a Japanese educational psychologist (Furukawa, 1927 & 1930; Nawata, 2014; Rogers et al., 2003; Sato et al., 1992; Wu et al., 2005). Yet, the epoch that most shape present-day research is a Japanese book (Nomi, 1971) written by Masahiko Nomi, a Japanese independent researcher, as well as an English book (Nomi et al., 1988) written by his son, Toshitaka Nomi, and Alexander Beshner (Nawata, 2014; Rogers et al., 2003; Sato et al., 1992).

Results of Recent Academic Studies

Questionnaire-based personality assessment, frequently used in psychology and consists of answering to multiple questions regarding multiple self-reported personality traits and integrating into several personality factors by statistical processing. In theory, this means that the self-reported answer will either directly or indirectly appear in the result. Although there are many academic studies on the relationship between blood type and personality, the inconsistency among results (Cattell et al., 1964; Cho et al., 2005; Cramer et al., 2002; Flegr et al., 2013; Furukawa, 1927 & 1930; Gupta, 1990; Jogawar, 1983; Kim et al., 2007; Lester et al., 1987; Mao et al., 1991; Nawata, 2014; Rogers et al., 2003; Sharifi et al., 2015; Sato et al., 1992; Shimizu et al., 2011; Wu et al., 2005) has led to the endless academic controversy about whether the relationship is scientifically confirmed. Many studies examined the association between blood type and personality using the "Big Five" personality test (Goldberg, 1990 & 1992), which has been extensively present in contemporary psychology. However, none of these results was certified as consistent (Cho et al., 2005; Cramer et al., 2002; Flegr, et al., 2013; Rogers et al., 2003; Sharifi et al., 2015; Shimizu et al., 2011; Wu et al., 2005). In 2014, a study that analyzed data of large-scale social surveys in Japan and the US was conducted, yet the study over 10,000 participants revealed no meaningful difference (Nawata, 2014).

Self-fulfilling Prophecy

Self-fulfilling prophecy refers to the phenomenon in which a person who believes in a prophecy learns to act in accordance with the prophecy, thereby bringing the prophecy to being. An example that psychologists have studied is astrology (Glick et al., 1986; Eysenck et al., 1982; Snyder, 1984). If a person's original personality and his/her sign matches, that tendency becomes stronger. Even if the personality and the sign do not initially match, the personality moves toward what has been indicated. In Japan, South Korea, and

Taiwan, roughly half of the people feel the relationship between blood type and personality is legitimate (Cho et al., 2005; Kamise et al., 1994 & 1995; Sato et al., 1992; Wu et al., 2005; Yamaoka, 2009). Logically, the self-fulfilling findings among astrology suggest that one's personality would change in a direction that fits relevant as personality description – of which shown by blood type. Based on this self-fulfilling prophecy hypothesis, several large-scale surveys in Japan which were designed or analyzed by psychologists or other academic researchers after 1990, clearly confirmed the phenomenon (Matsui, 1991; Muto et al., 2012; Ryu et al., 2007; Sakamoto et al., 2004; Yamaoka, 2009; Yamazaki et al., 1991). Archetypical sample sizes of these were 6,660 (Yamaoka, 2009), 11,766 (Matsui, 1991), 32,347 (Sakamoto et al., 2004; Yamazaki et al., 1991 – these two used the same sample) and over 100,000 (Muto et al., 2012; our estimation; the exact number was not specified in this report, although it alluded the size was much larger than preceding ones). Nevertheless, there is not existing coherent statistical difference in respondents without knowledge of astrology or blood type. The current scientific consensus is that these differences are self-fulfilling phenomena induced by the “contamination by knowledge” (Cramer et al., 2002; Eysenck, 1982; Kim et al., 2007; Matsui, 1991; Ryu et al., 2007; Sakamoto et al., 2004; Sato et al., 1992; Yamaoka, 2009; Yamazaki et al., 1991).

Contemporary Trends and Issues

After 2000, a growing number of studies proved the previously questioned link between blood type and physical constitution, with the exception being in the weak gastrointestinal tract: this demonstration proposed a new approach to medicine (Ewald et al., 2016; Risch, 2000). There had also been several studies on biological factors, which investigated whether physical constitution affected personality (Hobgood, 2011a & 2011b). In 2015, a genotype of blood type and the TCI Personality Test had been determined to be related, as predicted by blood type personality theory (Tsuchimine et al., 2015).

It is understood that personality involves complex interactions of genetic factors such as gender and age, and such interactions between these factors are typically non-linear (Kawamoto et al., 2015; Lehmann et al., 2013; Weisberg et al., 2011). However, the majority of the previous studies assumed that the effect of blood type is linear and coherent, regardless of gender or age (Cattell et al., 1964; Cho et al., 2005; Cramer et al., 2002; Flegr et al., 2013; Furukawa, 1927 & 1930; Gupta, 1990; Jogawar, 1983; Kim et al., 2007; Lester et al., 1987; Nawata, 2014; Rogers et al., 2003; Sharifi et al., 2015; Sato et al., 1992; Shimizu et al., 2011; Wu et al., 2005). Statistical methods used to design questionnaire-based personality tests, such as principal component analysis and correlation analysis, basically assume a linear relationship between variables. Therefore, there is no theoretical guarantee that these statistical methods or personality tests will accurately analyze real, non-linear data.

Methods

Participants

The data used in this study (Appendix A and Supplementary Material File) was collected by the means of crowdsourcing. With the intention of avoiding ethical issues, question items on blood type characteristics of this study were extracted from reviewed academic articles (Sato et al. 1992; Watanabe 1994). These items were checked again at the previously stated crowdsourcing company which has been passed the Japanese privacy mark (JIS Q 15001). The company confirmed that there was no problem with the question items. It provides customers with anonymized data and obtains informed consent from participants (respondents) prior to its surveys.

Survey 1 was conducted in 2018 as a preliminary study with a sample size of 1,000. Japanese single individuals with the age of 20-39 were asked to rate a total of 8 items representing 4 blood type (A, B, O, AB) characteristics, each with scores of 1-5 for their personality traits (the larger the number, the more fitting the trait), and scores of 1 to 4 of relationship and their knowledge on the level between blood type and

personality (the larger the number, the more related or informed). Survey 2 was conducted in 2019, and covered 2,000 Japanese people ages 20-59, further expanded blood type characteristics to 12 items, and asked the respondents to answer which blood type they thought these 12 items would categorize. In Survey 2, we added an item of marital status, since the age of participants spanned to 59, which meant that married persons were included.

Each participant's blood type of was determined by self-report, because most Japanese people know their blood type. In Survey 1, the dataset included only those who knew their blood type. Hence, the valid sample size was 1,000. In Survey 2, 141 out of all the 2,000 respondents, or 7.1% of the total, did not know their blood type. Thereby the data of the remaining 1,859 participants with known blood types were used.

The result of preliminary Survey 1 suggested that people who were less interested in blood type had traits of introvert, unsociable, and less interested in personality. Subsequently we added three new question items in Survey 2 to confirm this. These items were determined by the same process that was used to extract question items on the characteristics of the blood type. All samples were equally allocated by age of 10-year increments, and gender.

Procedure and Analytical Strategy

We deliberately selected the most suitable characteristics that would certainly produce the differences, and then focused our analysis on whether personality self-fulfillment was occurring or not: with data from participants who "have no knowledge of blood type characteristics" (hereinafter abbreviated as "no-knowledge group"). Survey 1 had only 120 "no-knowledge" participants therefore we would mainly use the 325 "no-knowledge" participants of Survey 2, who were free of "self-fulfilling", because they had no knowledge of blood type personality. Differences between groups with large and small characteristic differences according to blood type were also examined.

Our analytical methods were as follows:

Analysis 1: Cross-tabulation of blood type and personality - Survey 1 and 2

Analysis 2: Cross-tabulation of knowledge and personality -Survey 1 and 2

Analysis 3: Cross-tabulation of knowledge, extroversion and sociability - Survey 2 only

We set the alpha level to 0.05. The ANOVA, and the binominal test were used to analyze blood type characteristics. The effect sizes (Cohen, 1977) were also calculated. The dependent variables were analyzed by both the blood type factor and the covariates of gender, age (10-year increments), and marital status (since Survey 1 participants were all single, Survey 2 only). The distributions of blood types were almost equal to the Japanese average (Okubo, 1997), so the results of χ^2 test for the bias in both surveys were not significant.

Results

Analysis 1: Cross-tabulation of Blood Type and Personality

In Survey 1's ANOVA result (Table 1), 6 of the 8 items demonstrated the same results as those shown for blood type characteristics in the preceding psychology papers (Sato, 1992; Watanabe, 1994). Of these, 5 items were statistically significant at $p < 0.05$, or 4 items after the Bonferroni's correction. The magnitude of the difference: partial $\eta^2 = 0.030$ at the maximum, thus all the effect sizes were small.

Table 1. Survey 1 ANOVA for Characteristics Scores and Blood Type

Q1	Characteristics (Blood type)	Scores by blood type	Scores by blood type	Scores by blood type	Scores by blood type	F	Partial η^2	P
		A (N = 378)	B (N = 216)	O (N = 295)	AB (N = 111)			
a	Cooperative (A)	3.220 (1.191)	3.069 (1.246)	3.288 (1.164)	3.225 (1.093)	1.451	0.004	0.226
b	Open within peers (O)	3.251 (1.185)	3.343 (1.190)	3.349 (1.182)	3.459 (1.166)	1.030	0.003	0.378
c	Hard to be understood (AB)	3.259 (1.039)	3.389 (1.085)	3.132 (1.075)	3.550 (1.134)	5.004	0.015	0.002*
d	Self-paced (B)	3.188 (1.073)	3.407 (1.087)	3.153 (1.094)	3.360 (1.102)	3.106	0.010	0.026
e	Thoughtful & cautious (A)	3.619 (1.067)	3.403 (1.121)	3.454 (1.071)	3.523 (1.069)	2.256	0.007	0.080
f	Dual personality (AB)	2.913 (1.253)	3.162 (1.172)	2.854 (1.176)	3.486 (1.205)	9.403	0.025	< .001
g	Optimistic (B)	2.857 (1.208)	3.185 (1.213)	3.159 (1.237)	3.099 (1.191)	4.934	0.015	0.002*
h	Big-hearted (O)	2.899 (1.170)	3.037 (1.189)	3.383 (1.091)	3.045 (1.115)	10.210	0.030	< .001

Note. Standard deviations are in parentheses. Highest scores that match with blood type characteristics, and $p < 0.05$ are highlighted in bold. * $p < 0.05$ after the Bonferroni's correction.

On question item “Q2 Do you think blood type and personality are related?”, 8.3% of the respondents answered “strongly related”, 40.6% answered “somewhat related”, 24.6% answered “a little related”, 18.0% answered “not related at all” and 8.5% answered “I don’t know”. On question item “Q3 Do you know the characteristics and compatibility of blood types?”, 9.9% answered “I know well”, 54.6% answered “I know some”, 23.5% answered “I know a little” and 12.0% answered “I don’t know at all.”

In Survey 2's ANOVA result (Table 2), 11 of the 12 items gave the respondents' image of blood type and were statistically significant, or 9 items after the Bonferroni's correction. The magnitude of the difference: partial $\eta^2 = 0.053$ at the maximum and 0.019 at the average (the average of Survey 1 and 2 combined was 0.017), thus, the effect sizes were intermediate between small and medium.

Table 2. Survey 2 ANOVA for Characteristics Scores and Blood Type

Q1	Characteristics (Blood type)	Blood type	Blood type	Blood type	Blood type	F	Partial η^2	p
		A (N = 692)	B (N = 434)	O (N = 553)	AB (N = 180)			
a	Meticulous (A)	3.522 (0.998)	3.203 (1.055)	3.047 (1.059)	3.228 (1.118)	22.833	0.035	< .001

Q1	Characteristics (Blood type)	Blood type	Blood type	Blood type	Blood type	F	Partial η^2	p
b	Hard to be understood (AB)	3.130 (0.999)	3.311 (1.061)	3.123 (1.003)	3.544 (1.021)	10.764	0.017	< .001
c	Self-paced (B)	3.462 (1.007)	3.873 (0.952)	3.510 (0.978)	3.561 (1.037)	17.118	0.027	< .001
d	Positive (O)	2.903 (1.070)	2.993 (1.126)	3.074 (1.053)	2.922 (1.146)	2.746	0.004	0.042
e	Dual personality (AB)	2.448 (1.086)	2.599 (1.113)	2.409 (1.088)	3.322 (1.122)	35.047	0.053	< .001
f	Big-hearted (O)	3.012 (0.990)	3.025 (1.018)	3.228 (0.998)	2.972 (1.054)	6.163	0.010	< .001
g	Self-centered (B)	3.013 (1.023)	3.276 (1.086)	3.043 (0.992)	3.167 (1.075)	6.753	0.011	< .001
h	Serious (A)	3.671 (0.923)	3.537 (1.044)	3.486 (0.982)	3.656 (0.959)	4.360	0.007	0.005
i	Cheerful (O)	3.132 (1.043)	3.051 (1.018)	3.061 (1.011)	3.067 (1.050)	0.764	0.001	0.514
j	Loose (O)	3.156 (1.082)	3.288 (1.067)	3.608 (1.003)	3.261 (1.069)	20.706	0.031	< .001
k	Genius-like (AB)	2.23 (1.035)	2.311 (1.091)	2.204 (1.034)	2.639 (1.102)	8.836	0.014	< .001
l	Nervous (A)	3.516 (1.000.)	3.249 (1.058)	3.246 (1.050)	3.456 (1.037)	9.668	0.015	< .001

Note. Standard deviations are in parentheses. Highest scores that match with blood type characteristics, and $p < 0.05$ are highlighted in bold. * $p < 0.05$ after the Bonferroni's correction.

On question item “Q3 Do you think blood type and personality are related?”, 6.1% of the respondents answered “strongly related”, 35.7% answered “somewhat related”, 28.0% answered “a little related”, 20.1% answered “not related at all”, and 10.1% answered “I don’t know”. On question item “Q4 Do you know the characteristics and compatibility of blood types?”, 4.4% answered “I know well”, 41.4% answered “I know some”, 36.1% answered “I know a little”, and 17.5% answered “I don’t know at all.”

Analysis 2: Cross-tabulation of Knowledge and Personality

For each level of knowledge on blood type and personality, we calculated the corresponding simple averages of the scores answering to “fit to my personality” (Tables 3 and 4, and Appendix B).

Table 3. Survey 1 Scores of “Fit to My Personality” by Knowledge Level

Blood type	Knowledge level	Knowledge level	Knowledge level	Knowledge level
	1 (N = 120)	2 (N = 235)	3 (N = 546)	4 (N = 99)
My blood type	3.062	3.209	3.469	3.800
Other blood type	2.986	3.046	3.246	3.443

Note. The higher the level, the more informed.

Table 4. Survey 2 Scores of “Fit to My Personality” by Knowledge Level

Blood type	Knowledge level	Knowledge level	Knowledge level	Knowledge level
	1 (N = 325)	2 (N = 682)	3 (N = 770)	4 (N = 82)
My blood type	3.047	3.282	3.474	3.930
Other blood type	2.893	3.017	3.154	3.350

Note. The higher the level, the more informed.

Regardless of whether a person had knowledge of the characteristics of his/her blood type, the scores of characteristics for items that fit his/her blood type were higher than those of the other blood types. In both studies, the scores of characteristics which “did not fit” to one’s blood type in the “I know well” or “I know some” group were higher than those of characteristics “fit” to one’s blood type in the “I don’t know at all” (no-knowledge) group. In the no-knowledge group, scores of 4 out of 8 items were reversed in Survey 1, and 2 items out of all 12 were so in Survey 2. Thus, these items of Survey 2 were statistically significant in the binomial distribution at $p = 0.019$.

Analysis 3: Cross-tabulation of Knowledge, Extroversion and Sociability

According to the results obtained by a Japanese psychologist (Yamaoka, 2009), groups with more knowledge of blood type tended to be more interested in personality and other people. To confirm this, we investigated correlations by adding 3 items (Q1-m, n and o) in Survey 2. All 3 items were significant at $p < 0.001$ like Analysis 2; those with more knowledge were more likely to be extravert and sociable (Table 5).

Table 5. Survey 2 Result of Cross-tabulation by Knowledge Level, Extroversion and Sociability

Q1	Characteristics	Knowledge level	Knowledge level	Knowledge level	Knowledge level	F	p
		1 (N = 325)	2 (N = 682)	3 (N = 770)	4 (N = 82)		
m	Expressive and delicate	2.818 (1.040)	3.101 (0.978)	3.404 (0.983)	3.756 (1.192)	37.397	<.001
n	Friendly and sociable	2.403 (1.089)	2.740 (1.060)	2.936 (1.122)	3.305 (1.376)	24.479	<.001
o	Good and humane	2.705 (1.003)	2.977 (0.915)	3.152 (0.937)	3.585 (1.314)	26.627	<.001

Note. Standard deviations are in parentheses. The higher the level, the more informed. $p < 0.05$ are highlighted in bold.

Discussion

Issues with Personality Tests

Out of all 20 blood type characteristic items, 17 of them (14 after the Bonferroni’s correction) clearly demonstrated the differences as predicted. This study measured self-reported personality, and since many Japanese believe in the relationship between blood type and personality, conventional personality psychology

theory suggests the differences among the blood types will certainly appear, as prior mentioned. For the same reason, personality psychology hardly explains that no difference in blood type, or inconsistent results, appeared theoretically (Cattell et al., 1964; Cho et al., 2005; Cramer et al., 2002; Flegr et al., 2013; Furukawa, 1927 & 1930; Gupta, 1990; Jogawar, 1983; Kim et al., 2007; Lester et al., 1987; Mao et al., 1991; Nawata, 2014; Rogers, et al. 2003; Sharifi, et al. 2015; Sato et al., 1992; Shimizu et al., 2011; Wu et al., 2005).

Sung Il Ryu et al. reanalyzed So Hyun Cho et al's result of the Big Five test (Cho et al. 2005) and found statistically significant differences which match with blood type characteristics in 10 individual items (Ryu et al., 2007). This means that in the case of a "personality factor" composed of multiple items, the difference by the blood type decreases - few significant differences appear.

According to ANOVA results of this study (Appendix C), some items were influenced by age, gender, etc. than blood type, therefore consistent results cannot be obtained without considering these elements in a personality factor. This corresponds to the inconsistent results of many preceding studies conducted by psychologists.

In addition, the reason why psychological personality tests did not show differences in blood type (Cattell et al., 1964; Cho et al., 2005; Cramer et al., 2002; Flegr et al., 2013; Furukawa, 1927 & 1930; Gupta, 1990; Jogawar, 1983; Kim et al., 2007; Lester et al., 1987; Mao et al., 1991; Nawata, 2014; Rogers et al., 2003; Sharifi et al., 2015; Sato et al., 1992; Shimizu et al., 2011; Wu et al., 2005) was probably due to the halo effect (Kamise et al., 1994). Differences by the blood type are heavily depended upon "original" words. In many cases, differences did not appear if "similar" words were selected (Sakamoto et al., 2004; Yamaoka, 2009; Yamazaki et al., 1991). For example, Analyses 2 and 3 in Survey 2, respondents identified "cheerful", "expressive and delicate", "friendly and sociable", and "good and humane" in Q1 as characteristics of type O or A, but their actual answers did not yield any statistically significant differences. It should be noted that these items were expected to procure differences, since they were chosen as typical characteristics of blood type in previous surveys by psychologists (Watanabe, 1994).

Consistency with Preceding Studies

Preceding research by psychologists had concluded that there appeared no difference in personality among blood types, and that even if there was a difference predicted by the blood type, it was assumed to be the result of self-fulfillment prophecy phenomena (Kim et al., 2007; Sakamoto et al., 2004; Yamaoka, 2009; Yamazaki et al., 1991). However, backed on this study's data, it was highly likely that these phenomena occurred because gender and age were not taken into account. The experimental blood type prediction by AI (Amazon Machine Learning) in this study found that adding non-blood type variables, such as gender or age, to the training data considerably increased the accuracy (Appendix D). Moreover, when performing blood type prediction, AI sometimes failed to build its machine learning models, if gender or age of the training and prediction data were different. In this respect, AI technology may suggest that factors such as gender and age affect the characteristics of blood type. Therefore, gender, age, and other factors may offer a better explanation, even if past data were inconsistent.

The personality factors of the Big Five personality test are designed primarily to maximize the variances of respondents (Goldberg, 1990 & 1992). In general, the effect of blood type is smaller than that of gender or age (Nawata, 2014; Yamazaki et al., 1991). For this reason, differences in blood type, which appear in a single item, decrease significantly in personality factors if the gender or the age of the sample is not appropriately controlled. According to a study which compared the effect of blood type and age, the latter was approximately 4 times larger than the former (Yamazaki et al., 1991). Common statistical tests, such as the F -test, compare the variances of the raw data squared, so that the actual values compared are 4 squared, or 16 times. Thus, statistical tests are less likely to show significance. Given these large differences of influence and the non-linear nature of the actual effects, it can be very difficult to detect the effect of blood type with these personality tests. This offers an explanation to the inconsistent results (or no difference) in the actual Big Five personality test. The current personality psychology theory can explain the phenomenon

without contradictions.

Individual Differences in Personality Sensitivity

Taste and visual sensitivities vary among individuals, thereby sensory test (e.g., ISO 20613) considers this a self-evident assumption. By the same reasoning, personality sensitivity may vary from person to person, but psychologists rarely take this into account when designing common personality tests, such as the Big Five test (Goldberg, 1990 & 1992).

In Analysis 3, the average score of characteristics that did not fit their blood type in the “I know well”, “I know some” and “I know a little” groups had higher scores (on Survey 1, 3.443, 3.246 and 3.046, respectively) than the average score 2.986 of “no-knowledge” group. Survey 2 showed the same results. If the personality sensitivities were the same among these groups, the average score for non-blood-type-matching characteristics would be the same in each group, but this was not the case. In addition, the average scores of characteristics that “did not fit” his/her blood type in the “I know well” and “I know some” groups were higher than the average score for characteristics that “fit” his/her blood type in the no-knowledge group (3.062).

To explain the results obtained in this study, it is natural to attribute the small differences in the blood type between the two no-knowledge groups, not to the conventionally assumed self-fulfilling prophecy, but to their low sensitivity to their personalities. In Analysis 3, the no-knowledge groups displayed a tendency to be little interested in others and personality. The two groups seemed to unsociable, compared with the other groups, which was also consistent with the results of this study.

Occupation by Blood Type

Hans Eysenck and David Nias suggested that researchers should focus on occupations people select, rather than subjective self-assessments, to avoid the “contamination by knowledge” that arises in the case of astrology. They cited several examples in their book. However, the results were generally in negative (Eysenck et al., 1982).

The situation of blood type and personality is different, unlike astrology. Several academic studies had examined the relationship between blood type and occupation; significant differences were reported especially among politicians and athletes. For example, Masao Ohmura et al, a Japanese psychologist group said, “many Japanese prime ministers are type O” - statistically significant (Ohmura et al., 2013). Another example: Japan has many type O and AB foreign ministers and O defense ministers, which demonstrates statistical significance. In general, athletes tend to be type O. An Italian study said that type O are good athletes of long-distance running (Lippi et al., 2017). In both Japan and South Korea, type O is the best batter in professional baseball. Type B, considered to be at his/her own pace, is supposedly a better player in individual competition than in group. In Japan, there are far fewer B’s among athletes of the FIFA World Cup and female artistic swimmers, which are group games. On the contrary, B is prominent among the talented Japanese professional baseball batters and the professional golfers. Thus, in occupations, there lies distinct statistically significant differences predicted by blood type. These can be recognized as evidence that blood type is related to physical constitution and personality (Kanazawa, 2019).

Conclusion

We demonstrated a clear relationship between blood type and the self-reported personality of many single question items, which matches traits previously stated. The effect of blood type on personality was shown as partial $\eta^2 = 0.053$ at the maximum in this study - the effect size was medium. The same traits were observed in the two no-knowledge groups, though the difference was smaller. People with no-knowledge of blood type traits might be less concerned or sensitive to personality. A person without blood type knowledge had lower scores of sociability and empathy, while those with more knowledge had higher. If no-knowledge

group has low sensitivity of the personality, thereby the difference becomes small. This idea corresponds with the actual data.

Blood type does not uniquely affect a person's personality. On the contrary, various factors such as gender and age have an intricate and significant impact. This makes it impossible to achieve consistent results without taking these non-linear interactions into account. Traditional personality tests, such as the Big Five test, typically use a personality factor that consists of multiple items. In this case, the effect of blood type fluctuates, which can produce inconsistent results unless the gender, age, or other conditions of the sample are properly controlled. The explanation above is in good agreement with the theory of personality psychology. Our findings provide a new, if hypothetical, framework of how genes affect human personality.

Meanwhile, the sample in this study was limited to Japanese populations only, the AI training data was small in sample size, its use was experimental, and the interaction between gender and age is still unknown. Additional research using larger, global dataset is needed in order to address true implications as well as to improve algorithms and methodologies.

Appendix A

Question Items of Survey 1

Participants: 1,000 persons

(single men and women in his/her 20s or 30s, age and gender are equally allocated)

Q1. Please answer the following eight items about your personality.

Answer options

1: It fits to me not at all 2: It fits to me little 3: Cannot say either 4: It fits to me somewhat 5: It fits to me well

Question items

a. I am cooperative b. I am open in peers c. Others are hard to understand my personality d. I am self-paced and hardly affected by others e. I am thoughtful and take a cautious attitude toward others f. I have an uneven mood and sometimes it looks like a dual personality g. I am optimistic h. I am big-hearted

Q2. Do you think blood type and personality are related?

1: Not related at all 2: A little related 3: Somewhat related 4: Strongly related 5: I don't know

Q3. Do you know the characteristics and the compatibilities of blood types?

Answer options 1: I don't know at all 2: I know a little 3: I know some 4: I know well

Q4. Lastly, please tell us your blood type.

1: Type A 2: Type B 3: Type O 4: Type AB 5: I don't know

Q5. Gender (based on respondent registration information)

1: Male 2: Female

Q6. Age (depending on respondent registration information)

1: Teen or under 2: 20s 3: 30s 4: 40s 5: 50s 6: 60s 7: 70s or over

Question Items of Survey 2

Participants: 2,000 persons

(men and women in his/her 20s to 50s, age and gender are equally allocated)

Q1. Please answer the following eight items on your personality.

Answer options

1: It fits to me not at all 2: It fits to me little 3: Cannot say either 4: It fits to me somewhat 5: It fits to me well

Question items

a. Meticulous b. Hard to understand by others c. Self-paced d. Positive e. Dual personality f. Big-hearted g. Self-centered h. Serious i. Cheerful j. Loose k. Genius-like l. Nervous The following items m - o are ones to check whether subjects are extroverted or sociable, rather than examining personality differences by blood types (little difference among blood types in previous studies) m. Expressive and delicate n. Friendly and sociable o. Good and humane.

Q2. For the 15 items in Question 1, please choose one or more blood type(s) that you think will apply to your image.

1: Type A 2: Type B 3: Type O 4: Type AB

Q3. Do you think blood type and personality are related?

1: Not related at all 2: A little related 3: Somewhat related 4: Strongly related 5: I don't know

Q4. Do you know the characteristics and the compatibility of blood types?

Answer options 1: I don't know at all 2: I know a little 3: I know some 4: I know well

Q5. Lastly, please tell us your blood type.

1: Type A 2: Type B 3: Type O 4: Type AB 5: I don't know

Q6. Gender (based on respondent registration information)

1: Male 2: Female

Q7. Age (depending on respondent registration information)

1: Teen or under 2: 20s 3: 30s 4: 40s 5: 50s 6: 60s 7: 70s or over

Appendix B

Survey 1 Scores of “Fit to My Personality” by Knowledge Level

Simple average	Simple average	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)
My blood type	My blood type	1 (120)	2 (235)	3 (546)	4 (995)
Other blood type	Other blood type	3.062	3.209	3.469	3.800
Difference	Difference	2.986	3.046	3.246	3.443
<i>No. of reverse item(s)</i>	<i>No. of reverse item(s)</i>	0.075	0.163	0.222	0.358
Individual items	Individual items	4	3	1	1
		Knowledge level (N)	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)
Q1-a	A	1 (120)	2 (235)	3 (546)	4 (995)
		2.830	3.110	3.308	3.531

Simple average	Simple average	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)
Q1-b	Others	<i>2.863</i>	2.984	<i>3.352</i>	3.329
	Difference	<i>-0.033</i>	0.126	<i>-0.044</i>	0.202
	O	<i>2.861</i>	<i>3.000</i>	3.526	3.720
Q1-c	Others	<i>2.972</i>	<i>3.038</i>	3.522	3.560
	Difference	<i>-0.111</i>	<i>-0.038</i>	0.005	0.160
	AB	<i>2.923</i>	3.412	3.630	4.050
Q1-d	Others	<i>3.298</i>	3.245	3.207	3.547
	Difference	<i>-0.375</i>	0.167	0.423	0.508
	B	3.333	3.404	3.364	3.708
Q1-e	Others	3.052	3.097	3.289	3.493
	Difference	0.281	0.307	0.074	0.218
	A	3.489	<i>3.374</i>	3.707	3.938
Q1-f	Others	3.311	<i>3.441</i>	3.467	3.638
	Difference	0.179	<i>-0.067</i>	0.239	0.302
	AB	3.154	3.382	3.413	4.113
Q1-g	Others	3.051	2.847	2.973	3.180
	Difference	0.103	0.535	0.440	0.931
	B	3.208	<i>2.830</i>	3.264	<i>3.450</i>
Q1-h	Others	2.498	<i>2.860</i>	3.132	<i>3.560</i>
	Difference	0.711	<i>-0.030</i>	0.132	<i>-0.100</i>
	O	<i>2.694</i>	3.159	3.538	3.880
	Others	<i>2.848</i>	2.858	3.028	3.230
	Difference	<i>-0.153</i>	0.301	0.510	0.650

Survey 2 Scores of “Fit to My Personality” by Knowledge Level

Simple average	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)
	1 (325)	2 (682)	3 (770)	4 (82)
My blood type	3.047	3.282	3.474	3.930
Other Blood type	2.893	3.017	3.154	3.350
Difference	0.154	0.265	0.319	0.581
<i>No. of reverse item(s)</i>	<i>1</i>	<i>2</i>	<i>1</i>	<i>1</i>

Individual items	Individual items	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)
Q1-a		1 (325)	2 (682)	3 (770)	4 (82)
	A	3.314	3.430	3.626	3.941
	Others	2.991	3.146	3.196	3.558
Q1-b	Difference	0.323	0.284	0.430	0.383
	AB	3.188	3.554	3.587	4.500
	Others	3.157	3.098	3.256	3.389
Q1-c	Difference	0.031	0.456	0.331	1.111
	B	3.519	3.770	4.095	4.174
	Others	3.423	3.444	3.587	3.686
Q1-d	Difference	0.096	0.326	0.508	0.488
	O	2.878	<i>2.949</i>	3.235	3.765
	Others	2.734	<i>2.954</i>	3.000	3.066

Individual items	Individual items	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)	Knowledge level (N)
Q1-e	Difference	0.144	<i>-0.005</i>	0.235	0.699
	AB	2.781	3.108	3.627	4.375
	Others	2.265	2.436	2.580	2.882
Q1-f	Difference	0.516	0.672	1.047	1.493
	O	3.061	3.097	3.176	3.941
	Others	2.787	2.957	3.130	3.046
Q1-g	Difference	0.274	0.140	0.046	0.895
	B	<i>2.896</i>	3.297	3.367	3.739
	Others	<i>3.006</i>	2.957	3.161	3.502
Q1-h	Difference	<i>-0.110</i>	0.340	0.206	0.237
	A	3.322	3.604	3.816	<i>4.029</i>
	Others	3.296	3.548	3.633	<i>4.089</i>
Q1-i	Difference	0.026	0.056	0.183	<i>-0.060</i>
	O	2.714	<i>2.940</i>	3.529	4.000
	Others	2.689	<i>3.052</i>	3.226	3.501
Q1-j	Difference	0.025	<i>-0.112</i>	0.303	0.499
	O	3.347	3.493	<i>3.265</i>	4.000
	Others	3.087	3.152	<i>3.364</i>	3.326
Q1-k	Difference	0.260	0.341	<i>-0.099</i>	0.674
	AB	2.188	2.615	2.827	2.875
	Others	2.055	2.206	2.317	2.727
Q1-l	Difference	0.133	0.409	0.510	0.148
	A	3.356	3.528	3.534	3.824
	Others	3.223	3.252	3.402	3.424
	Difference	0.133	0.276	0.132	0.400

Appendix C

Survey 1 ANOVA Result of Each Elements

Items	Blood type	Blood type	Gender	Gender	Age	Age
	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>
Q1-a (A)	1.451	0.226	0.000	0.992	0.038	0.846
Q1-b (O)	1.030	0.378	0.958	0.328	8.481	0.004
Q1-c (AB)	5.004	0.002	0.466	0.495	0.003	0.959
Q1-d (B)	3.106	0.026	0.443	0.506	5.874	0.016
Q1-e (A)	2.256	0.080	0.591	0.442	0.223	0.637
Q1-f (AB)	9.403	< .001	0.269	0.604	8.031	0.005
Q1-g (B)	4.934	0.002	2.392	0.122	4.927	0.027
Q1-h (O)	10.210	< .001	6.947	0.009	2.391	0.122

Survey 2 ANOVA Result of Each Elementsn

Items	Blood type	Blood type	Gender	Gender	Marital status	Marital status	Age	Age
	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>

Items	Blood type	Blood type	Gender	Gender	Marital status	Marital status	Age	Age
Q1-a (A)	22.833	< .001	10.262	0.001	10.603	0.001	3.322	0.069
Q1-b (AB)	10.764	< .001	4.865	0.028	11.109	< .001	0.445	0.505
Q1-c (B)	17.118	< .001	2.440	0.118	11.644	< .001	8.144	0.004
Q1-d (O)	2.746	0.042	6.439	0.011	1.318	0.251	8.385	0.004
Q1-e (AB)	35.047	< .001	3.102	0.078	0.671	0.413	13.718	< .001
Q1-f (O)	6.163	< .001	1.357	0.244	0.982	0.322	1.649	0.199
Q1-g (B)	6.753	< .001	0.369	0.544	2.631	0.105	29.476	< .001
Q1-h (A)	4.360	0.005	6.921	0.009	4.524	0.034	1.503	0.220
Q1-i (O)	0.764	0.514	23.603	< .001	18.090	< .001	0.070	0.792
Q1-j (O)	20.706	< .001	49.138	< .001	0.001	0.979	65.148	< .001
Q1-k (AB)	8.836	< .001	55.963	< .001	2.111	0.146	7.823	0.005
q1-l (A)	9.668	< .001	1.555	0.213	0.487	0.485	0.664	0.415

Appendix D

Experimental Result of Blood Type Prediction Using AI

All the data of survey 2 (excluding 141 participants who did not know their blood type) were stored in Amazon S3, and with Amazon Machine Learning, 12 characteristics, including gender, age, and marital status, were used as training data for prediction targeting for the blood type. Multinomial logistic regression algorithm was chosen for the prediction. We divided the whole data into five groups of same sample size. Each group was estimated as the prediction data, the rest four groups as the training data, and then the average of the five predictions was calculated. In these cases, since the sample sizes of the AI training data were small (this means that the prediction errors might become larger if we used the raw data of 1-year increment of age), a dummy variable of 10-year increments was used [20s = 2, 30s = 3, 40s = 4, 50s = 5]. The accuracy rates were 45.8% ($F1 = 0.367$) in the group that had good knowledge of blood type characteristics (542 participants with scores equal to 3 or higher in both item Q3 (relation) and item Q4 (knowledge)), and 40.1% ($F1 = 0.281$) in the entire 1,859 participants. When gender, age, and marital status were excluded from both the learning and training data, the accuracy rates fell to 42.3% ($F1 = 0.334$), and 39.6% ($F1 = 0.274$) respectively. The most common blood type among Japanese is type A, which accounts for 39.1%, of the total population (Okubo, 1997). Hence, the accuracy rate became 39.1% if all the participants were assumed to be the type. Amazon Machine Learning predicted the blood type at a higher accuracy than this value in all cases (however, only the 45.8% accuracy was significant at $p = 0.032$). For all 20 individual predictions, 17 were more accurate than this value, and the binomial test was significant at $p = 0.011$. In addition, considering that all F1-scores were higher than chance ($1/4 = 0.25$), Amazon's AI predicted human blood types with a higher probability than chance.

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Supplementary Material File (Dataset).xlsx available at <https://authorea.com/users/718274/articles/703530-abo-blood-type-and-personality-traits-evidence-from-large-scale-surveys-in-japan>